

Epic 3: Frontend Development

1. Frontend Technology Stack Choice

The presentation layer of EduGenie uses a clean, light architecture designed for rapid page loading and clear UI layouts. Instead of relying on complex, heavy client-side JavaScript frameworks like React or Vue, the project combines server-side rendered **HTML5**, **CSS3**, and **Jinja2** templates. This design approach delivers three primary operational benefits:

Zero Build Steps: Front-end changes require no compilation, asset bundling, or node module packaging, keeping deployment simple.

Minimal Client Overhead: HTML is fully rendered on the host server before transmission, ensuring instant, accessible views on low-powered mobile devices.

Clean State Mapping: Form fields, text inputs, and previous prompt values map directly back into form controls during post-request views, creating a smooth, single-page application experience without complex state stores.

2. User Interface Screen Layout and Experience Structure

The entire application interface is contained within a single clean web layout. The interface uses a centered, card-based structural wrapper that keeps content easy to read on both desktop and mobile viewports.

Header Section: Features a prominent title banner with an illustrative emoji visual anchor alongside a subtitle that clearly communicates the multi-model purpose of the application.

Interactive Form Controls: Contains an intuitive drop-down selection menu where users choose their target task type. Each option clearly indicates its backend AI model (e.g., Local LaMini-Flan-T5 vs Gemini 3.8 flash) so users understand where their queries are processed. A vertical text area underneath provides space for pasting large blocks of study material or typing open-ended questions.

Dynamic Response Block: This card uses conditional rendering rules and appears only after the server successfully processes a request. It features a prominent visual badge that identifies the specific backend model used. The output text is styled with a light blue background and a solid border accent, using CSS formatting rules like `white-space: pre-wrap` to accurately preserve the markdown spacing, paragraphs, and list layouts generated by the AI modules.